

INDIAN INSTITUTE OF TECHNOLOGY, ROPAR



CP302 - CAPSTONE PROJECT - I

REPORT

On

**Radiological Risk Evaluation of Fly Ash
Utilization in Asphalt Pavements**

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CERTIFICATE

This is to certify that the *Capstone Project-1* titled **“Radiological Risk Evaluation of Fly Ash Utilization in Asphalt Pavements”** prepared by *Desireddy Sreelakshmi and Tanishq* has been reviewed and approved for the Interim Evaluation.

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DECLARATION

We , hereby, declare that the report titled “***Radiological Risk Evaluation of Fly Ash Utilization in Asphalt Pavements***” has been carried out by us under the supervision of *Prof. Dr. Surya Kant Sahdeo*, Department of Civil Engineering.

This report represents my original work, and all sources of information used have been properly cited and acknowledged.

We also declare that this work has not been submitted previously for any other academic requirement.

Date: 12/05/2026

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I. INTRODUCTION

Road construction requires large quantities of natural aggregates, leading to depletion of natural resources and environmental concerns. At the same time, industrial waste products such as **fly ash**, resulting from thermal power plants, also need to be disposed of properly. The utilization of fly ash in pavement construction is also a sustainable solution, reducing industrial waste and the consumption of natural aggregates.

Fly ash is also used for **stabilizing pavement** and soil construction due to the fine particle size of the fly ash. It is added to the pavement construction mix to provide better **particle packing, workability**, and **strength** to the pavement mixture by filling the voids between the aggregates and improving the bonding within the asphalt mixture. This helps to provide **better stability and durability** to the pavement mix, and fly ash is considered an alternative construction material for pavement construction.

However, fly ash is also reported to contain **naturally occurring radioactive materials (NORM)** such as **Radium-226 (Ra-226)**, **Thorium-232 (Th-232)**, and **Potassium-40 (K-40)**. Thus, when fly ash is added to pavement construction mix, the radiological safety of the construction mix needs to be evaluated to ensure that the radiation exposure is within the permissible limits for the construction workers and the general population.

In the current research, **Bituminous Concrete Grade-2 (BC-2)** mix is prepared by replacing natural aggregates with fly ash at different replacement levels of **10%, 20%, and 30%**. First, a conventional control mix is prepared without fly ash to understand the properties of the bitumen mixture. Then, fly ash is added at different replacement levels, and the mixes are tested for different parameters.

The **Marshall Stability Test** is performed to study the strength, stability, and deformation characteristics of the bitumen mixes. **Gamma Spectrometry** is performed to check for Ra-226, Th-232, and K-40, which are radioactive materials, to ensure the radiological safety of the mixes. Additionally, **Scanning Electron Microscope (SEM)** test is performed to study the microstructure of fly ash and its interaction with the bitumen mixes, giving an insight into how fly ash affects the internal structure of the bitumen mixes.

The results of the conventional control mix and fly ash mixes are compared to determine the **safe and effective proportion of fly ash**, which can be used as a replacement material in BC-2 bitumen mixes.



2. OBJECTIVES

The main objectives of this project are:

1. To determine the **activity concentrations** of Radium-226, Thorium-232, and Potassium-40 in fly ash using **gamma spectrometry**.
2. To calculate radiation hazard parameters such as **Radium Equivalent Activity (Raeq)**, **External Hazard Index (Hex)**, **Annual Gonadal Dose Equivalent (AGDE)**, and **Outdoor Absorbed Dose Rate**.
3. To analyze the microstructural characteristics of fly ash using **Scanning Electron Microscopy**.
4. To assess the **radiological safety** and **suitability** of fly ash for outdoor construction applications.



3. LITERATURE REVIEW

Several studies have been conducted to evaluate the **radioactivity and radiological safety of construction materials containing fly ash** using gamma spectrometry.

- A study by **Mehra et al. (2019)** investigated the variation of natural radionuclides such as **Radium-226, Thorium-232, and Potassium-40** in cement when mixed with different percentages of fly ash. The activity concentrations were measured using a **NaI(Tl) gamma spectrometer**, and radiological hazard parameters such as **radium equivalent activity, external hazard index, gamma index, and annual effective dose** were evaluated. The study reported that the **concentration of radionuclides increases with increasing fly ash content**, and it was recommended that the **fly ash content should be limited to about 20% to maintain safe radiation levels** in building materials.
- Another study focused on determining the **radioactivity of natural raw materials used in construction** through gamma spectrometric analysis. The researchers measured the specific activities of radionuclides and calculated parameters such as **effective specific activity, radium equivalent activity, and radiation hazard indices** to evaluate the safety of construction materials. The results indicated that if the **effective specific activity remains below 370 Bq/kg**, the materials can be safely used in construction without significant radiological risk.

- Previous research has also examined the **radioactivity of fly ash generated from coal combustion** and its potential impact when used in construction materials. Studies showed that fly ash contains natural radionuclides originating from coal, and therefore the radiological properties of materials containing fly ash must be assessed using **gamma spectrometry and radiological hazard indices** before their application in construction.
- Although most of the previous studies focused on **cement-fly ash mixtures or raw construction materials**, the present study differs by analyzing the **compacted Marshall bituminous specimens** after grinding them into powder. The radionuclide concentrations obtained from gamma spectrometry will be used to calculate radiological hazard parameters such as **radium equivalent activity and dose indices**. This approach helps evaluate the **radiological safety of fly ash when used in bituminous pavement materials**, which has been less explored in earlier studies.



4. MATERIALS USED

4.1 Aggregates

Aggregates were used as the main component of the bituminous mixture. They were tested for their physical properties before being used in the preparation of Marshall specimens.

4.2 Bitumen

Bitumen (VG-30) was used as the binding material in the asphalt mix. A bitumen content of **5.5% by weight of the total mix** was used for the preparation of Marshall test specimens.

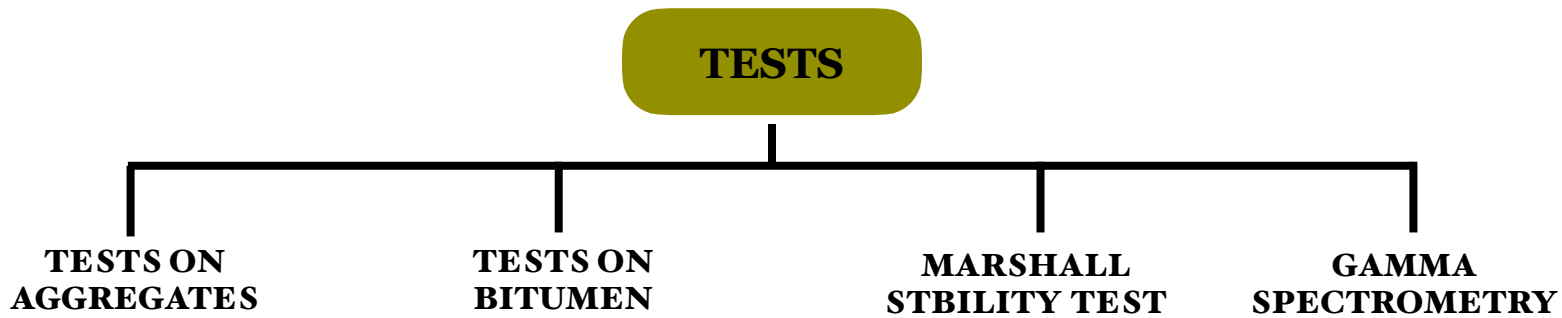
4.3 Fly Ash

Fly ash was used to study its potential application in bituminous mixtures and its radiological characteristics. The samples were analyzed to determine the activity concentrations of Radium-226, Thorium-232, and Potassium-40.



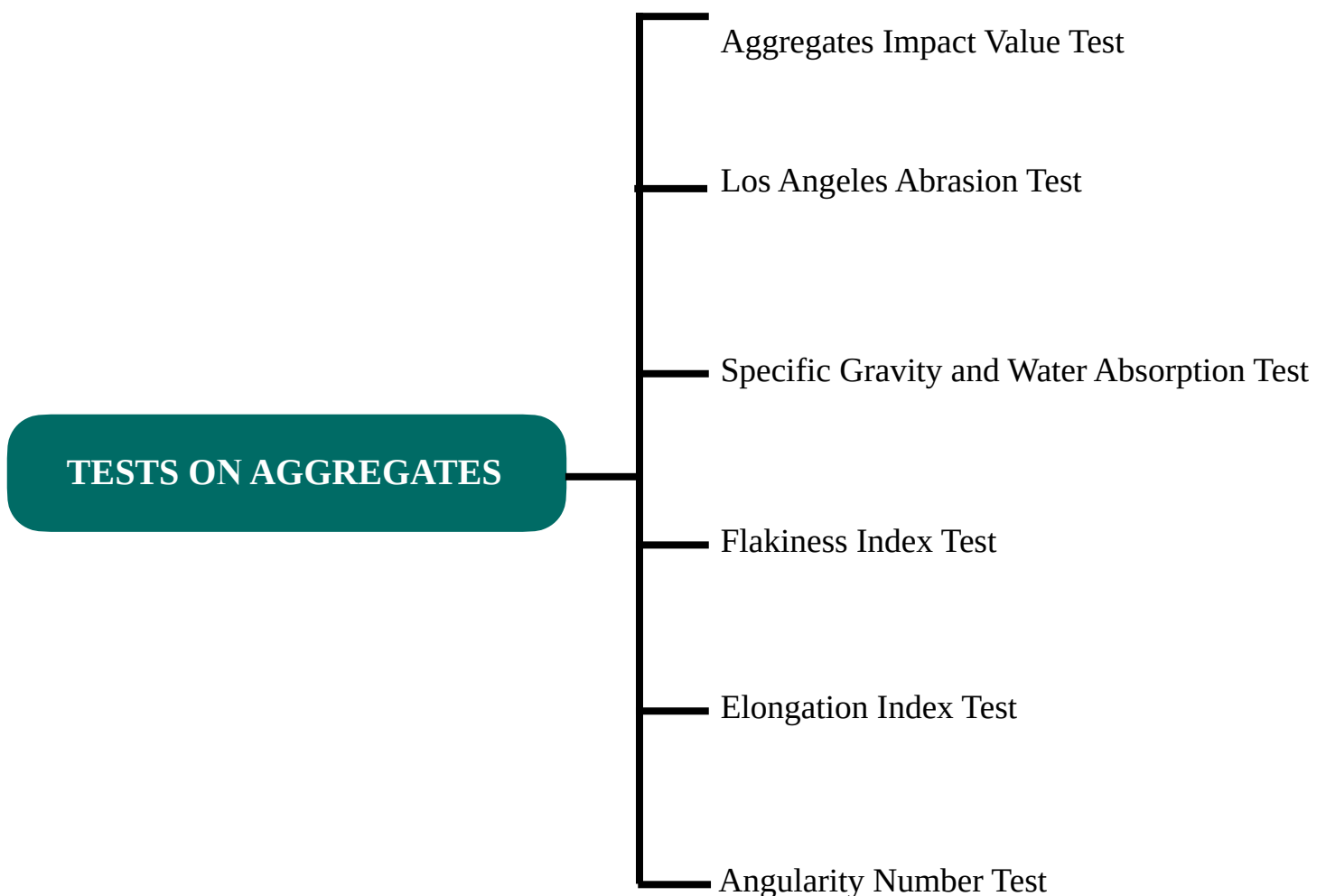
5. METHODOLOGY

The methodology involved testing aggregates and bitumen to evaluate their suitability for asphalt mixtures. Marshall specimens were prepared and tested to determine the stability and flow characteristics of the mix. Fly ash samples were also prepared and later analyzed using gamma spectrometry to determine radionuclide activity concentrations.



5.1. Tests on Aggregates

- The following tests were performed on aggregates:



5.1.1 Aggregates Impact Value Test

(IRC: 37 , IS 2386 Part (IV): 1963)

- This test is used to determine the toughness of aggregates and their resistance to sudden impact or shock, which is important for pavement materials subjected to traffic loads.
- From the experiment:
Obtained Aggregate Impact Value = **22.67 %**
Permissible Value as per IRC:37 = **30 %**
- Since the obtained value **22.67 % is less than 30 %**, the aggregates satisfy the IRC requirements and are **suitable for pavement construction**.

5.1.2. Los Angeles Abrasion Test (IRC: 37 , IS 2386 Part (IV): 1963)

- This test is conducted to determine the resistance of aggregates to abrasion and wear caused by traffic movement.
- From the experiment:
Obtained Los Angeles Abrasion Value = **27.9 %**
Permissible Value as per IRC:37 = **30 %**
- Since the obtained value **27.9 % is within the permissible limit**, the aggregates have **adequate resistance to abrasion and are suitable for pavement construction**.

5.1.3. Specific Gravity and Water Absorption Test (IS 2386 Part (III): 1963)

- The **Aggregate Specific Gravity Test** is performed to determine the density characteristics and water absorption capacity of aggregates, which influence durability and strength.
- From the experiment:
Specific Gravity = **2.57**
Apparent Specific Gravity = **2.70**
Water Absorption = **1.855 %**
Standard values as per IS 2386:
- Typical Specific Gravity Range = **2.5 – 3.0**
Permissible Water Absorption < **2 %**
- Since the obtained values fall within the recommended limits, the aggregates are **suitable for use in pavement construction**.

5.1.4. Flakiness Index Test (IRC : 109, IS 2386 Part (I): 1963)

- This test determines the percentage of flaky particles in aggregates, which may reduce the strength of the pavement structure.
- From the experiment:
Flakiness Index = **8.86 %**
Permissible Value as per IRC:109 = **25 %**
- Since **8.86 % is well below the permissible limit**, the aggregates **meet the required specification**.

5.1.5. Elongation Index Test (IRC : 109, IS 2386 Part (I): 1963)

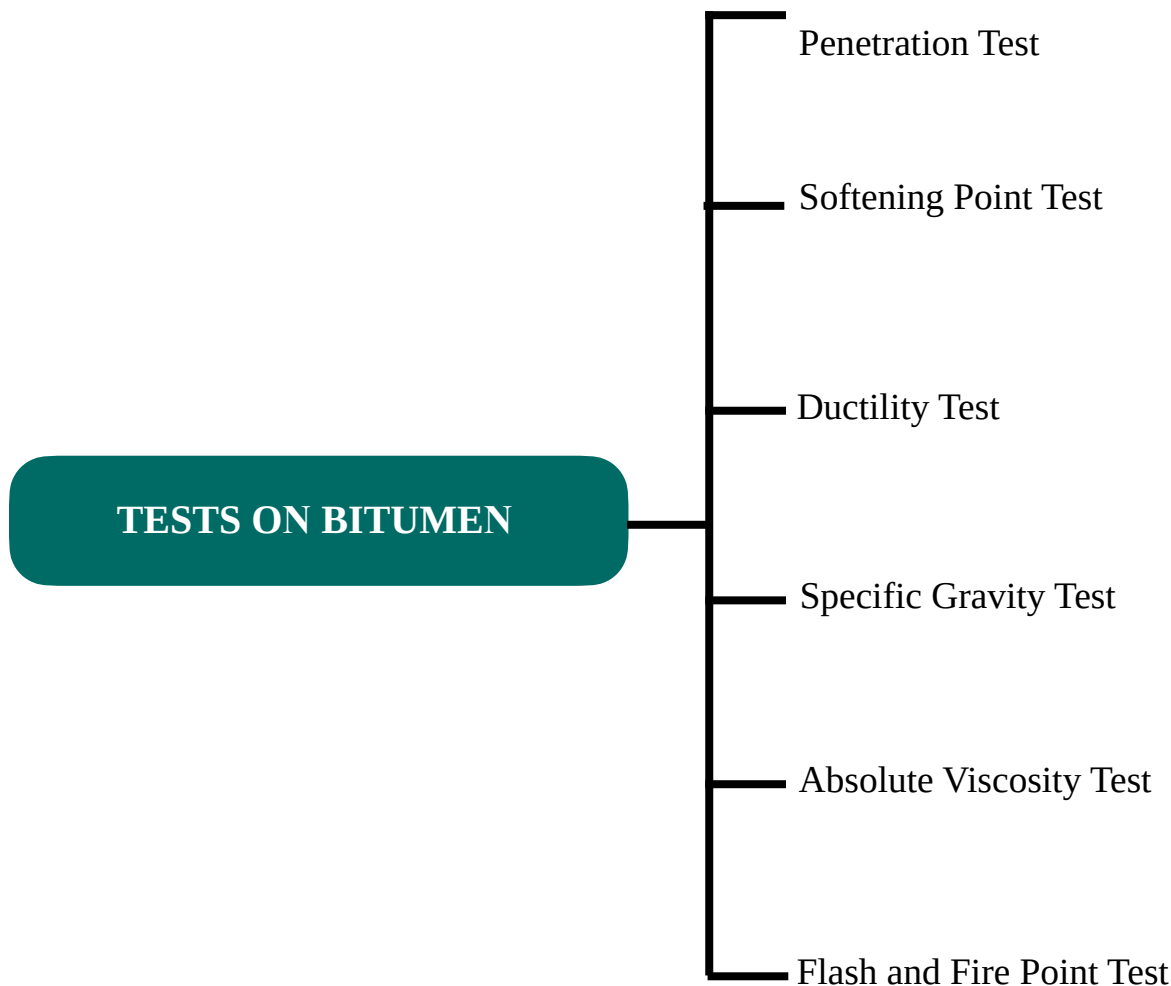
- This test is used to determine the percentage of elongated particles in aggregates, which may affect compaction and stability.
- From the experiment:
Elongation Index = **3.67 %**
Permissible Value as per IRC:109 = **25 %**
- Since **3.67 % is within the permissible limit**, the aggregates **satisfy the specification requirements**.

5.1.6. Angularity Number Test (IS 2386 Part (I): 1963)

- This test determines the angularity and interlocking characteristics of aggregates, which influence the stability of bituminous mixtures.
- From the experiment:
Angularity Number = **4.48**
Typical Range for Semi-Rounded Aggregates = **2 – 6**
- Since the obtained value **4.48 lies within this range**, the aggregates are **semi-rounded and suitable for pavement applications**.

5.2. Tests on Bitumen

- The following tests were performed on bitumen :



5.2.1. Penetration Test (IS 1203: 1978 , IS 73 :2013)

- This test is used to determine the consistency and hardness of bitumen by measuring the depth to which a standard needle penetrates the sample under specified conditions.
- From the experiment:
Mean Penetration Value = **35.48 (0.1 mm)**
Permissible Range for VG-30 bitumen as per IS 73 = **30 – 40 (0.1 mm)**
- Since the obtained value **35.48 lies within the permissible range**, the bitumen corresponds to **VG-30 grade** and **satisfies the specification for pavement applications**.

5.2.2. Softening Point Test (IS 1205 : 1978)

- The **Ring and Ball Softening Point Test** determines the temperature at which bitumen softens and begins to flow under standard conditions.
- From the experiment:
Mean Softening Point = **55.125 °C**
Permissible Range for VG-30 bitumen as per IS 73 = **47 °C – 65 °C**
- Since the obtained value **falls within the specified range**, the bitumen **meets the standard requirement**.

5.2.3. Ductility Test (IS 1208 : 1978)

- This test measures the ability of bitumen to stretch without breaking, which indicates good adhesion and flexibility.
- From the experiment:
Average Ductility Value = **86 cm**
Permissible Value as per IS 73 > **75 cm**
- Since the obtained value **86 cm is greater than the minimum requirement**, the bitumen **possesses adequate ductility and is suitable for pavement construction**.

5.2.4. Specific Gravity Test (IS 1202 : 1978)

- This test determines the density of bitumen relative to water, which is important for mix design calculations.
- From the experiment:
Specific Gravity = **1.025**
Typical Range for Paving Bitumen = **1.01 – 1.06**
- Since the obtained value **lies within this range**, the bitumen **satisfies the standard requirements**.

5.2.5. Absolute Viscosity Test (IS 1206 (Part II) : 1978)

- The **Bitumen Absolute Viscosity Test** is conducted to determine the resistance of bitumen to flow at a specified temperature, which influences the workability and performance of the mix.
- From the experiment:
Mean Viscosity = **3246 seconds**
- This value corresponds to the viscosity characteristics of **VG-30 grade bitumen**, indicating **adequate resistance to flow**.

5.2.6. Flash and Fire Point Test (IS 1209 : 1978)

- The **Flash and Fire Point Test** is conducted to determine the temperature at which bitumen vapors ignite momentarily (flash point) and continue burning (fire point).
- From the experiment:
Flash Point = **344 °C**
Fire Point = **350 °C**
Minimum Flash Point as per IS 73 = **220 °C**
- Since the obtained values **are much higher than the minimum requirement**, the bitumen is **safe for heating and mixing operations**.

5.3 Marshall Stability Test

The **Marshall Stability Test** was carried out to evaluate the strength and deformation characteristics of the bituminous mixture with and without the addition of fly ash.

5.3.1. Sample Preparation

- Aggregates were dried and heated to the required mixing temperature.
- Bitumen was heated separately to achieve proper fluidity for mixing.
- Heated aggregates were mixed thoroughly with bitumen at a constant **bitumen content of 5.5% by weight of the total mix.**



Fig : Mixing Bitumen and Aggregates

- Initially, **three Marshall specimens were prepared without fly ash** to obtain the control mix results.
- In the next stage, fly ash will be incorporated as a filler material in proportions of **10%, 20%, and 30%**, while keeping the bitumen content constant.
- For each fly ash percentage, **three specimens will be prepared.**
- The prepared mixture was placed in a **standard Marshall mould** and compacted using a Marshall compaction hammer to produce cylindrical specimens.



Fig 2 : Prepared Samples



Fig 3: Compacting the mix

5.3.2. Testing Procedure

1. After compaction, the specimens were removed from the mould and allowed to cool at room temperature.
2. The specimens were conditioned in a **water bath maintained at the specified temperature** before testing.
3. Each specimen was placed in the **Marshall stability testing machine**.
4. Load was applied at a constant rate until failure occurred.
5. During the test, the **maximum load carried by the specimen (Marshall Stability)** was recorded.
6. The **corresponding deformation at the point of failure (Flow Value)** was also recorded.



Fig 4: Removing specimen from mould

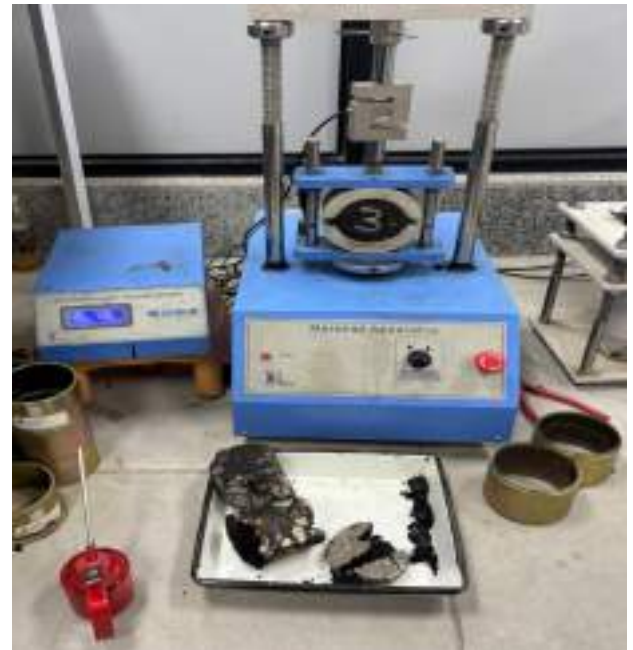


Fig 5 : Marshall Stability Test

5.3.3. Aggregate Gradation for Marshall Mix

- The aggregate gradation used for preparing Marshall specimens was selected according to the **BC-2 gradation limits for bituminous concrete**. Aggregates were sieved through standard sieve sizes and combined in suitable proportions to achieve the required gradation. A total of **1200 g of aggregates** was used for the mix, and the weight of aggregates corresponding to each sieve fraction was calculated based on the specified gradation. The Marshall specimens were prepared for the conventional mix **without fly ash** using a constant **bitumen content of 5.5%**. The total weight of the bituminous mix was **1266g**, consisting of **1200 g aggregates and 66 g bitumen**.



Fig 6 : Gradation of aggregates

Sieve Size	% Cumulative by weight of total	% Cumulative	% Retained	Weight in grams
19	100	100	0	0
12.5	90-100	95	5	60
10	70-88	79	16	192
4.75	53-71	62	17	204
2.36	42-58	50	12	144
1.18	34-48	40	10	120
0.6	26-38	30	10	120
0.3	18-28	22	8	96
0.15	12-20	15	7	84
0.075	4-10	7	8	96
Pan	0.00	0	7	84
Sum			100	1200

Table -1: Gradation of Marshall BC-2

For VG-30, as per MoRTH: 5013, the Table-1 should be followed.

5.3.4. Observations and Calculations

- The weight of the specimen in air, in water, and in saturated surface dry condition was measured.
- Using these values, the **bulk specific gravity of each specimen** was calculated.
- The **Marshall stability value** and **flow value** for each specimen were recorded from the testing machine.
- The **average stability and flow values** were calculated for the specimens.
- These results were used to evaluate the **performance of the bituminous mix with different fly ash proportions while keeping the bitumen content constant**.
- The detailed observations and calculated values obtained from the experiment are presented in the following tables.

	Sample No.	Weight in air in gms (A)	Mass under water in gms (C)	SSD mass (B)	Bulk Specific Gravity $G_{mb} = A / (B - C)$
	1	1216	693	1216.5	2.322
Conventional	2	1240	701	1244	2.283
	3	1250	711	1251.5	2.312
	1	1234	678.5	1234	2.222
Flyash 10% by weight	2	1255	688.5	1254	2.219
	3	1243.5	667	1244	2.155
	1	1244	646	1244	2.080
Flyash 20% by weight	2	1240	640	1240	2.066
	3	1228	623	1228	2.128

Table -2: Specific Gravity Calculations

- Based on the **Marshall correction factor table (ASTM / MORTH)** and specimen heights, the corresponding correction factors are:

	Sample No.	Height (mm)	Correction Factor
	1	63	1.00
Conventional	2	67	0.96
	3	65	0.96
	1	72	0.92
Flyash 10% by weight	2	70	0.94
	3	70	0.94
	1	75	0.89
Flyash 20% by weight	2	74	0.90
	3	75	0.89

Table -3: Correction Factors

Correction for Marshall Stability Value

- Since the Marshall specimen height may vary slightly from the standard height of **63.5 mm**, the measured stability values are corrected using standard correction factors. These correction factors depend on the thickness (height) of the specimen and are taken from the standard Marshall correction factor table. The corrected stability value is calculated using the following relation:

$$\text{Corrected Stability} = \text{Measured Stability} \times \text{Correction Factor}$$

	Sample No.	Stability Value (kN)	Corrected Stability Value (kN)	Flow (mm)
Conventional	1	22.05	22.05	3.80
	2	21.79	20.91	4.78
	3	27.59	26.48	3.98
Flyash 10% by weight	1	22.54	20.74	3.71
	2	26.01	24.45	2.27
	3	24.32	22.86	2.91
Flyash 20% by weight	1	24.28	21.61	3.40
	2	25.26	22.73	2.04
	3	20.97	18.66	3.19

Table -4: Stability Value and Flow corresponding to Impact Rate of 50mm/min



Fig 7 : Stability Value and Flow value

5.3.5. Result and Discussion

	Conventional Mix	Flyash 10% by wt.	Flyash 20% by wt.	Standard value
Average bulk specific Gravity	2.30	2.19	2.09	-
Average Stability Value (kN)	23.12	24.29	23.5	Min. 9
Average Flow Value (mm)	4.19	2.963	2.876	2 - 4

Table 5: Avg. Bulk specific gravity, stability value and flow value

- From the comparison of Marshall Stability and Flow values with IRC specifications, it can be concluded that the fly ash modified mixes performed better than the conventional mix.
- Among all the mixes, the 10% fly ash mix showed the best overall performance with the highest stability value of 24.29 kN and an acceptable flow value of 2.963 mm within the IRC recommended range of 2–4 mm.
- The conventional mix exhibited a slightly higher flow value of 4.19 mm, indicating comparatively higher deformation under loading.
- Although the 20% fly ash mix also satisfied IRC requirements, its stability value was slightly lower than the 10% fly ash mix.
- Therefore, the incorporation of 10% fly ash can be considered the optimum replacement level, as it improves strength, stability, and resistance to deformation while satisfying standard specifications.

5.4. Gamma Spectrometry

Gamma spectrometry is a technique used to **detect and measure radioactive elements in a sample by analyzing the gamma rays emitted during radioactive decay**. It is commonly used to determine the activity concentrations of natural radionuclides such as Radium-226, Thorium-232, and Potassium-40 present in materials. In this study, it is used to analyze the **ground Marshall specimen** to evaluate its radiological characteristics.

5.4.1. Specimen Preparation For Gamma Spectrometry

1. After completing the Marshall stability test, the **compacted Marshall specimen (cylindrical sample)** was collected for radiological analysis.
2. The specimen was **crushed and ground into fine powder** to obtain a homogeneous material.
3. The powdered sample was then **sieved through a 150 μm sieve** to remove larger particles and ensure uniform particle size.
4. The processed sample was **placed in a clean airtight container and sealed properly** to prevent the escape of radon gas.
5. The sealed samples were **stored undisturbed for approximately 6 weeks (about 42 days)** before measurement.
6. This storage period allows **secular equilibrium between the parent radionuclide Radium-226 and its decay products such as Radon-222, Lead-214, and Bismuth-214**.
7. Since **Radon-222 has a half-life of about 3.82 days**, storing the sample for several half-lives (about 4–6 weeks) ensures radioactive equilibrium.
8. After equilibrium is achieved, the **sealed sample is placed in the gamma spectrometer** to determine the activity concentrations of natural radionuclides

5.4.2. Radiometric Analysis

- Gamma-ray measurements were performed using a **Canberra p-type High Purity Germanium (HPGe) detector (GC2018-10854)** coupled with a spectroscopic amplifier and 8K multichannel analyzer (MCA). The detector was enclosed in a lead shielding assembly to reduce background radiation and was maintained at low temperature using liquid nitrogen. The detector was biased to +2900V. Prior to sample analysis, the system was calibrated for energy over a range of 39.5 - 1408 keV using standard multi-gamma sources such as Europium-152 and Cobalt-60.

- Background spectra were recorded under identical conditions and subtracted from the sample spectra to obtain net peak areas. Each prepared sample was placed in a fixed geometry over the detector end-cap and counted for 12 hours to achieve good counting statistics.
- **MAESTRO** software was used to collect and readout the data. The collected data were analyzed offline using linux advanced multi parameter system (LAMPS) analysis software. The X-axis (horizontal) represents the gamma ray energy (channel number), whereas the Y-axis (vertical) represents the activity or number of counts of gamma photons of a particular energy.
- Radionuclide identification was based on characteristic gamma emissions of progeny such as Lead-214, Bismuth-214, Actinium-228, and Potassium-40.
- The activity concentration of each radionuclide (Bq/kg) was calculated using the relation:

$$A = N \epsilon \cdot P_{\gamma} \cdot t \cdot m$$

where A is the activity concentration (Bq/kg),

N is the net peak area (counts),

ϵ is the detector efficiency at the corresponding gamma energy,

P_{γ} is the emission probability of the gamma line,

t is the counting time (s), and

m is the mass of the sample (kg).

5.4.3. Radiation Hazard Parameters

- The radiological hazard of fly ash samples is assessed using standard radiation indices calculated from the activity concentrations of Radium-226, Thorium-232, and Potassium-40 obtained from gamma spectrometry analysis. The activity concentrations are represented as C_{Ra} , C_{Th} , and C_K respectively (in Bq/kg).
- The **Radium Equivalent Activity (R_{eq})**, which represents the combined radioactivity of these radionuclides, is calculated using:

$$R_{eq} = C_{Ra} + 1.43C_{Th} + 0.077C_K$$

- The **External Hazard Index (H_{ex})** is used to evaluate the radiation hazard from the material and is given by:

$$H_{ex} = 370C_{Ra} + 259C_{Th} + 4810C_K$$

- The **Annual Gonadal Dose Equivalent (AGDE)**, which estimates the radiation dose to sensitive organs, is calculated as:

$$AGDE = 3.09C_{Ra} + 4.18C_{Th} + 0.314C_K$$

- The **Outdoor Absorbed Dose Rate (D_{out})** due to gamma radiation from natural radionuclides is determined using:

$$D_{out} = 0.427C_{Ra} + 0.662C_{Th} + 0.0432C_K$$

- These parameters will be calculated using the activity concentrations obtained from gamma spectrometry to assess the **radiological safety of fly ash for outdoor construction applications**.



6. RESULTS

The results for spectrometric analysis for three samples of conventional mix:

S. No.	Sample ID	U-238 (Bq/kg)	Th-232 (Bq/kg)	K-40 (Bq/kg)
1	S1	42.36 ± 1.32	35.93 ± 1.29	441.27 ± 23.75
2	S2	40.9 ± 1.05	37.66 ± 1.39	430.16 ± 24.15
3	S3	40.53 ± 1.07	33.34 ± 1.38	367.47 ± 26.3
Average		41.26	35.64	412.97
Standard Deviation		0.97	2.17	39.79

Table -6: Activity Concentrations

- The spectrometric analysis shows the presence of U-238, Th-232, and K-40 in all conventional mix samples. Among them, K-40 exhibits the highest activity concentration, while U-238 and Th-232 show comparatively lower values.
- The low standard deviation values indicate good consistency among the samples, although K-40 shows relatively higher variation.
- Overall, the conventional mix shows moderate natural radioactivity, mainly dominated by K-40.

S. No.	Sample ID	Ra _{eq} (Bq/kg)	H _{ex}	H _{in}	D _{out} (nGy/h)
1	S1	127.72	0.34	0.017	60.94
2	S2	127.88	0.35	0.016	60.98
3	S3	116.50	0.31	0.017	55.25
Average		124.03	0.33	0.0167	59.06
Standard Deviation		6.52	0.02	0.00	3.29
Standard Values		< 370	< 1	< 1	< 55

Table -7: Hazard indices and radiological dose parameters in conventional mixture

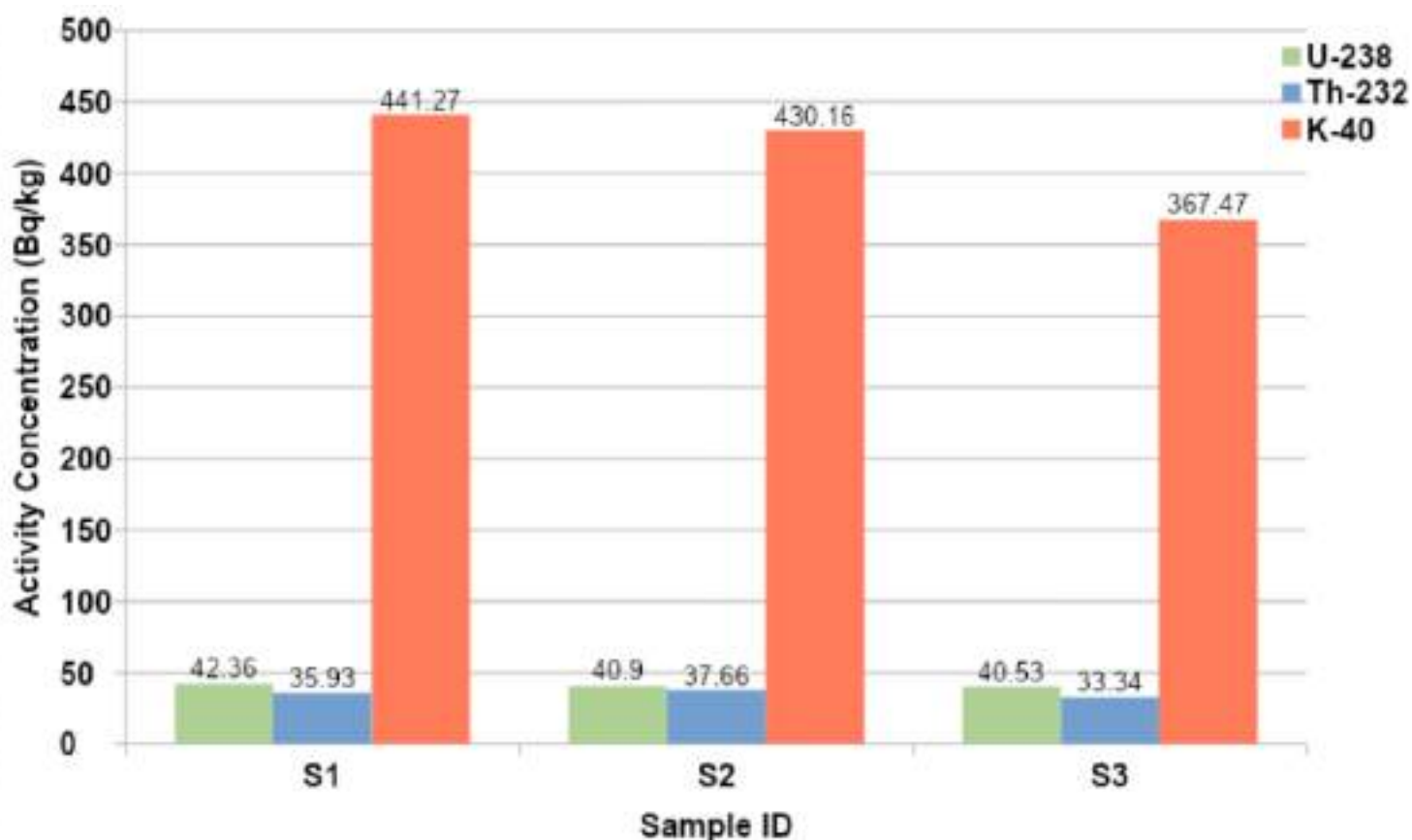


Fig. 7 : Bar chart representing activity concentrations

7. DISCUSSION

- The present study was carried out to evaluate the suitability of fly ash utilization in BC-2 bituminous pavement mixes from both mechanical and radiological perspectives. Marshall Stability tests were conducted on conventional and fly ash modified mixes prepared with 10% and 20% fly ash replacement while maintaining a constant bitumen content of 5.5%.
- The experimental results showed that the incorporation of fly ash improved the performance characteristics of the asphalt mixture. The 10% fly ash mix exhibited the highest Marshall stability value of 24.29 kN, which was higher than both the conventional mix (23.12 kN) and the 20% fly ash mix (23.5 kN). All the mixes satisfied the minimum IRC/MORTH requirement of 9 kN, indicating adequate load carrying capacity and structural stability for pavement applications.
- The flow values obtained for the fly ash modified mixes were within the IRC recommended range of 2–4 mm, whereas the conventional mix showed a slightly higher flow value of 4.19 mm, indicating comparatively higher deformation under loading. The reduction in flow values with the addition of fly ash indicates improved stiffness and resistance to plastic deformation. The bulk specific gravity decreased with increasing fly ash content due to the lower density of fly ash compared to conventional mineral filler materials. Overall, the Marshall test results indicate that the incorporation of fly ash enhances the mechanical performance of the bituminous mix, with 10% fly ash showing optimum performance.
- Gamma spectrometry analysis was carried out to evaluate the activity concentrations of U-238, Th-232, and K-40 in the conventional mix samples. The results showed that K-40 had the highest activity concentration among the radionuclides, while U-238 and Th-232 showed comparatively lower values. The calculated radiological hazard parameters such as Radium Equivalent Activity (R_{eq}), External Hazard Index (H_{ex}), Internal Hazard Index (H_{in}), and Outdoor Absorbed Dose Rate (D_{out}) were found to be within the recommended safety limits. The average R_{eq} value of 124.03 Bq/kg was significantly lower than the permissible limit of 370 Bq/kg, while the average H_{ex} and H_{in} values were much lower than unity, indicating negligible radiological hazard. Although the average outdoor absorbed dose rate was slightly higher than the world average value, the obtained results still indicate that the conventional mix does not pose significant radiation risk for outdoor pavement applications.

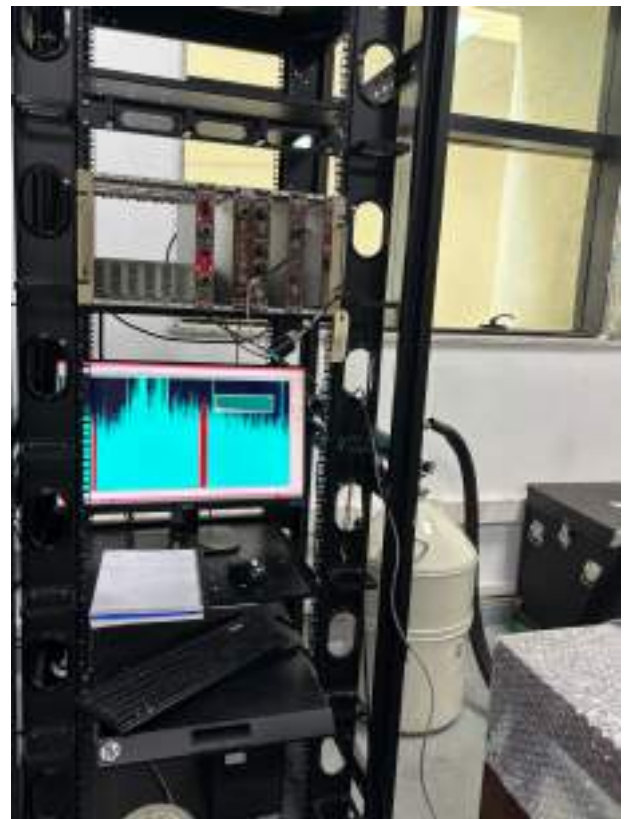


Fig.8 : Apparatus for Radiometric Analysis



8. CONCLUSION

- Based on the experimental investigation, it can be concluded that the use of fly ash in BC-2 bituminous mixes improves pavement performance while providing a sustainable method for industrial waste utilization. The aggregate and bitumen properties satisfied IRC and IS specifications, confirming their suitability for pavement construction.
- The Marshall Stability test results showed that all mixes satisfied IRC/MORTH requirements. Among them, the 10% fly ash mix exhibited the highest stability with acceptable flow values, indicating improved resistance to deformation and better overall performance.
- Gamma spectrometric analysis identified the presence of U-238, Th-232, and K-40, with K-40 showing the highest activity concentration. However, the calculated radiological hazard parameters remained within permissible safety limits, indicating no significant radiological risk.
- Overall, the study shows that fly ash can be safely and effectively used as a filler material in asphalt pavements, with 10% fly ash replacement being the optimum proportion for improved performance and radiological safety.

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